Summary of the corpus

Status on 13 October 2026

Telmo Miguel-Medina

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1. Identification

Searches run on 12 October 2026, with no date limit.

Source	Records
Scopus	1,380
Web of Science Core Collection	801
IEEE Xplore, query X1 (nodes and testability)	545
IEEE Xplore, query X2 (stimuli and test sets)	488
Forward citation tracking of the control set	692
Author searches, 2020 onwards	422
Total identified	**4,328**
Removed: no authors (front matter, proceedings, standards)	74
Removed: document type	7
Removed: duplicates	1,615
Corpus for screening	**2,632**

Of the 2,632 records, 1,882 come from the database searches and 750 only from citation and author tracking.

The search retrieves 23 of the 24 works of the control set. The remaining one (Stenbakken and Souders, 1987) has no DOI and will enter through backward citation tracking, which is pending: 99 references cited by five or more non-excluded records are not yet in the corpus.

2. Composition

The corpus spans 1963 to 2026.

Period	Corpus	First-pass “include”
Up to 1979	38	15
1980–1989	89	35
1990–1999	454	97
2000–2009	678	142
2010–2019	678	139
2020–2026	695	40
Total	**2,632**	**468**

About 1,480 records are conference papers and 1,065 are journal articles; the rest are reviews, book chapters and magazine articles. The 2020–2026 period is inflated by the author searches, which bring in other research lines of the same groups.

3. Screening: first pass

The figures in this section and the next come from the AI-assisted first pass on titles and abstracts. They are provisional: every decision will be checked by hand, and the counts will change.

First-pass decision	Records
Include	468
Doubt (goes to full text)	274
Exclude	1,890

Exclusion reason	Records
E4. Test technique, structure or tool without selection	622
E3. Other domain	572
E2. Diagnosis without selection	336
E1. Digital only	153
E5. Not a primary study	124
E7. Language	83

Up to 742 records may therefore need full-text assessment.

4. First look at the included studies

Pre-coding of the 468 first-pass inclusions, from the abstract only. It gives the order of magnitude, not results.

What is selected (not exclusive)	Studies
Stimuli	228
Test nodes	126
Measurements or specification tests	103

Level	Studies
Integrated circuit	148
Board	33
Not stated in the abstract	287

Objective	Studies
Fault isolation	163
Fault detection	153
Specification compliance	76
Parameter identification	76

Metaheuristics are the most frequent algorithm family that can be recognised from the abstract (57 studies), followed by sensitivity-based methods (38) and machine learning (33).

5. Full texts

Full-text retrieval has started with the studies of highest priority. Of the 468 first-pass inclusions, 22 PDFs are in hand and 43 more are open access.